

Empirical Research in Finance

11. Introduction to Academic Writing

Beatriz Mariano

Why write well — and what today gives you

Your result is only as good as your reader's ability to follow it. The reader reads **structure first**, numbers second.

By the end of today you should be able to:

- ① Organise a finance paper in its standard **components**.
- ② Turn a vague idea into a **one-line, testable hypothesis** with an expected sign.
- ③ Report a result as a **finance story** — direction, magnitude, meaning.
- ④ Build **stand-alone** tables/figures, and cite others' work correctly.

Structuring an argument

The typical components of an **empirical** research paper are:

- 1 Introduction
- 2 Hypothesis development
- 3 Data
- 4 Methodology
- 5 Results
- 6 Conclusion

Structure: Introduction

- What is the research question?
 - Put the main result in paragraph 2 or 3 (don't bury the lead!).
- Why is it important (to markets, or firms, or households, or regulators)?
- Preview of the approach and results in more detail.

Tip: It is often easier to write the introduction *after* everything else.

Structure: Related literature

- Either the final part of the introduction or a separate section.
- Group papers into **buckets** (topics / sub-topics), and contrast them:
 - Papers A, B, C examine *Topic 1* — A argues . . . , while B finds . . .
 - By contrast, papers D, E analyse *Topic 2* — D shows . . .
- **Do not list papers.** Build a map: what we know, what is debated, what is missing — and how you contribute.

Literature slide example: ESG and the cost of capital in Germany

Bucket 1: Theory / mechanism

- *Why* would ESG affect the cost of capital?
- Channels: risk, investor demand, regulation

Bucket 2: Evidence from debt markets

- ESG and loan terms: evidence is mixed
- ESG and spreads/yields: often lower

Bucket 3: Evidence from stock markets

- ESG and expected returns: mixed
- ESG and the investor base: shifts toward ESG funds

Bucket 4: The German / EU setting

- Bank-based finance + EU disclosure rules matter
- This is where **your** contribution sits

Structure: Hypothesis development

A hypothesis is a **precise** statement that is directly testable.

- *Not testable:*

- “Long-term lending relationships facilitate monitoring, hence relationships are beneficial for the borrower.”

- *Testable:*

- “Relationship loans, on average, carry a **lower** AISD (all-in spread drawn) than non-relationship loans.”

- Explain the economic intuition (the **why**) and competing arguments (in favour/against).

- It's okay if the data reject your hypothesis.

Turning an idea into a testable hypothesis

Example: Start with a story:

- “Firms with more analyst coverage disclose more information, so investors are less surprised.”

Translate into **measurable** objects:

- Outcome: earnings surprise
- Treatment/proxy: analyst coverage (# analysts)
- Controls: size, volatility, industry, time FE

Hypothesis (testable):

- H_1 : Higher analyst coverage is associated with **smaller** absolute earnings surprises.

Tip: Write the hypothesis in one line, with the expected sign.

The data section should:

- make your dataset **replicable**,
- make your sample choices **transparent** and the data easy to compare,
- make your results **credible** (no hidden data tricks).

What to include:

- Data sources and sample construction.
- Merges and identifiers (what links to what?).
- Key variable definitions (with intuition; details can go in a table).
- Descriptive statistics + comparison to prior work.

Adapt to method: a separate methodology section for special methods; for standard ones, group methodology and results by hypothesis.

Example — a simple regression. Do more leveraged firms invest less?

$$\text{CAPEX}_{i,t} = \alpha + \beta \text{Leverage}_{i,t-1} + \gamma' \mathbf{X}_{i,t-1} + \mu_i + \lambda_t + \varepsilon_{i,t}$$

- i = firm, t = year (or quarter)
- μ_i = firm fixed effects, λ_t = year fixed effects
- $\mathbf{X}$ = controls (size, profitability, cash holdings, Tobin's Q, etc.)

Coefficient of interest: $\beta < 0$ implies higher leverage is associated with lower investment (conditional on controls). Economic intuition: Consistent with debt overhang.

Structure: Results (answer the question)

Results are **answers**, not output. For the key coefficient in each table, give:

- **Direction** — up or down?
- **Magnitude** — how big, in **bps** / **%** / **euros**?
- **Meaning** — which hypotheses hold, and what do we learn about firms/investors?

Make it readable: highlight **one** coefficient (don't read the whole table); if the result is weak, say so and show what you tried next.

Economic significance: *"A one-SD increase in X is associated with a 30 bps change in spreads."*

Rewrite the result

✗ **Weak:** “The coefficient on leverage is negative and statistically significant ($t = -3.1$), confirming our hypothesis.”

What's wrong? No magnitude, no economic meaning, no reader takeaway.

✓ **Better:** “A one-standard-deviation increase in leverage is associated with a **1.8 percentage-point** lower investment rate — about **12%** of the sample mean. Consistent with H_1 , more-indebted firms invest less, in line with debt-overhang.”

- **Answer the question.** (1–2 sentences, no new results)
- **Give the number.** Main effect + economic magnitude (bps/%/euro)
- **Explain why.** Mechanism/ interpretation in plain language
- **Say why it matters.** Implications for investors, firms, policymakers
- **Briefly discuss.** Limitations + best next step with more time/data

Tables and figures

- Keep them in the **main text** (not hidden in the appendix).
- **Make them stand alone:**
 - understandable without reading the paragraph
 - informative captions (what, method)
 - readable variable names (e.g., Number of directors _{$t-1$} , not `dir_n_l1`)
- **Be precise:** units, scaling, and clear labels on rows/columns/axes.

Tip: **Stack models in one table:** put related specifications in separate columns.

Table example

Table IV
Determinants of Loan and Bond Credit Spreads

This table reports regressions of all-in-drawn loan spreads and bond swap spreads on firm, loan, and bond characteristics. Variables are defined in the Appendix. Loans with missing data for *Secured Loan* are assumed to be unsecured in this table. Within R^2 represents the goodness of fit after accounting for month fixed effects (but not rating or industry effects). t -statistics based on standard errors clustered by firm and month are reported in brackets. * and ** denote p -values less than 0.05 and 0.01, respectively.

Log(Swap Spread)	Loans			Bonds		
	(1)	(2)	(3)	(4)	(5)	(6)
Quasi-market leverage	0.40** [2.68]	0.25 [1.58]	0.36* [2.31]	1.20** [5.63]	0.99** [4.80]	1.21** [5.07]
Asset volatility	0.18 [0.79]	0.03 [0.10]	0.29 [1.16]	1.19** [3.64]	0.84* [2.20]	0.92* [2.06]
Maturity	0.05** [3.19]	0.03* [1.99]	0.01 [0.78]	0.06** [3.64]	0.07** [3.91]	0.10** [5.00]
Log(Loan amount)		-0.05** [-4.38]	-0.03** [-2.86]		0.01 [0.75]	0.03 [1.21]
Log(Quasi-market assets)		0.01 [0.47]	0.004 [0.23]		-0.14** [-7.05]	-0.13** [-5.68]
Profitability		-1.25* [-2.16]	-0.69 [-1.09]		-2.65** [-3.37]	-2.65** [-3.11]
Asset market-to-book		-0.01 [-0.43]	-0.003 [-0.09]		-0.04 [-1.28]	0.02 [0.51]
Secured loan		0.15** [2.80]	0.14** [3.73]		0.03 [0.58]	-0.003 [-0.05]
Term loan		0.14** [4.71]	0.15** [6.00]		-0.01 [-0.36]	0.02 [0.61]
Bank debt/total			-0.01 [-0.14]			-0.17 [-1.48]
Month FEs	X	X	X	X	X	X
S&P rating FEs	X	X	X	X	X	X
Two-digit SIC FEs		X	X		X	X
Adj. R^2	0.80	0.82	0.86	0.79	0.82	0.84
Within R^2	0.70	0.74	0.79	0.72	0.77	0.80
Observations	1,633	1,633	1,093	1,633	1,633	1,093

How to write it: pick *one* row — e.g. “Holding other firm and loan characteristics fixed, a one-unit increase in quasi-market leverage raises the loan spread by about 36%.” Then say why.

Source: Schwert, M., 2020. Does borrowing from banks cost more than borrowing from the market? *The Journal of Finance*, 75(2), pp.905–947.

Acknowledging the work of others

Any idea that is not yours must be clearly labelled as such.

In the text:

- Hanley (1993) argues that . . .
- IPO underpricing may be related to demand discovery (Hanley 1993).
- If two or more authors: Chava and Roberts (2003) or Smith et al. (2008).
- When continuing: “the author” or “this paper”.

In the reference list (alphabetical order):

References

Hanley, K. W. (1993). “The underpricing of initial public offerings and the partial adjustment phenomenon.” *Journal of Financial Economics*, 34, 231–250.

Often, the exact format is less important than consistency.

Quoting, paraphrasing, and AI tools

- **Quote** (rare): exact words in quotation marks + citation + page.
- **Paraphrase** (usual): your own words — *still* needs a citation.
- Copying without attribution is **plagiarism**, even if reworded loosely.

AI writing tools:

- Fine for grammar, structure, and clarity — you are responsible for every claim and number.
- **Not** a source: it can invent citations and “facts”. Verify everything against the original paper.
- Follow the course / HU rules on disclosure.

- The term paper is an **exam**.
 - I can only answer clarification questions.
- Submit your report as a PDF via Moodle.
- Presentation matters: structure, writing style, and clean tables/figures.
- Writing takes time — start early and keep it clear, precise, and concise.

Application to Bachelor Thesis Seminar

- Apply by end of March/September: send application form + transcript to finance-group@hu-berlin.de. You can apply before this seminar's grade is available.
- Supervisor = BTS teacher.
- BTS: hybrid, individual work, follows HU term dates.
- BTS includes: topic choice, data collection, short proposal presentation (last 4 weeks in-person).
- After BTS: official registration + $\approx$ 60 days to write.
- 12 ECTS.

Formal requirements — Application form — Moodle file

Examples of Past Theses

- Borrowing Under Stress: The Effect of the COVID-19 Crisis on Bond Spreads
- Choosing a Debt Path: A Comparative Study of Corporate Loans and Bonds
- Are Dirty Assets Affecting Returns? The Pollution Premium in European Markets
- Bank Lending and the Real Estate Market: a Survey
- Gender Gap at the Executive Level: Taste-Based or Statistical Discrimination?